RELATIVE WEAKLY ASSOCIATED AND ASSOCIATED POINTS AGREE FOR A LOCALLY FINITE TYPE MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 0-2000 granted to this lane by CONTROL 8.83.

Editorial note. The proof is short: it reduces the relative statement to the absolute one fibre by fibre and names that input; nothing is omitted and the reduction is the content of the card.

Editorial note. The single reference of the counted spans is recorded as a non-load-bearing navigation pointer; the argument that consumes it stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- Associated and weakly associated points agree on locally Noetherian schemes. Lemma lemma-ass-weakly-ass (700-712), used at proof lines 1220-1222. Is the absolute statement applied on each fibre.

Source reference. Original lemma 05AR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6859–6859. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes which is locally of finite type. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Then $WeakAss_{X/S}(\mathcal{F}) = Ass_{X/S}(\mathcal{F})$.*

Proof. This is true because the fibres of f are locally Noetherian schemes, and associated and weakly associated points agree on locally Noetherian schemes, see Lemma 05AR. □

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.